

Caleb Liu

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EDUCATION & COURSEWORK

Rensselaer Polytechnic Institute, School of Computer Science and Business – Troy, NY **May 2029**
Bachelor of Science in Computer Science and Business Analytics Minor: Finance Cumulative GPA: 3.64/4.00

Computer Science: Computer Science Principles, Data Structures, Rensselaer Center for Open Source

Math: Intro to Linear Algebra

Fall 2026: Foundations of Computer Science, Computer Architecture and Operating Systems, Mathematical Foundations of Machine Learning

TECHNICAL SKILLS

[Next.js](#), Python, MongoDB – [Node.js](#), Tailwind, [React.js](#), Java, PyTorch, C++, C#, Vue, SQL, Javascript

RELEVANT EXPERIENCE

Project Volition | Machine Learning Researcher | Troy, NY **September 2025 – Present**

- Developed 100% of graph neural network in **PyTorch** to predict brain wave data and write data gathering procedures
- Implemented 3 feature extraction functions: **Intrinsic Model**, **Hilbert Huang** transforms, and **Hjorth parameters**

HackRPI | Director of Technology | Troy, NY **December 2025 – Present**

- Managed 3-developer team in 3 **SCRUM** cycles to build open source **Next.js** website and design UI
- Updated website domains and forged relations with other tech clubs, delegating and marketing web game development

HackRPI | Junior Director of Technology | Troy, NY **September 2025 – December 2025**

- Overhauled website compatibility with mobile screen sizes using [Next.js](#), TypeScript, **Tailwind** CSS, [React.js](#)
- Increased visual accessibility by 230% using HTML, CSS, JavaScript, ensuring element visibility while eliminating overlap
- Created realistic 3D game and models using [Three.js](#) and **Blender**, and custom-built tools for rig animation and thematic lights

MoreNBest Decor | Automation Lead | Columbia, SC **March 2023 – September 2025**

- Expanded automation coverage by 90%, automating fetching inventory reports, analyzing sales, email content analyzation
- Implemented temporal dynamic customer pricing using inventory reports for an Amazon seller using **Amazon Seller Central**
- Scraped data using **GmailAPI**, **Selenium**, **SMTP**, **IMAP**, analyzing email senders, titles, contents, and files attached
- Replaced 100% of virtual communication between seller and distributor with autonomous email updates

PROJECTS

Plantilir | HackRPI | Troy, NY **November 2025**

- Scrapes internet and creates information profiles on people, gathering personal and career info and info about social network
- Maps 2d to 3d space, allowing for touchless interactions in [Three.js](#) using **mediapipe** to recognize hand and facial geometries

Large Language Model | Personal | Troy, NY **September 2025 – October 2025**

- Tokenizes Moby Dick vocabulary into vectors using **NLTK** library and **PyTorch**
- Predicts sentences and answers questions using a Feed Forward Neural Network with multiple layers

Voice Assistant | Personal | Troy, NY **September 2025 – October 2025**

- Listens and responds to user voice commands using python Speech Recognition Library
- Searches Youtube for specific music and tracks music preferences using WebBrowser API and Python
- Searches Wikipedia or Google and summarizes specific information using Wikipedia API and WebView API

NapCast | Cornell BigRedHacks | Ithaca, NY **September 2025**

- Generates podcasts using a **PyTorch** model trained with scraped Spotify podcasts on **Google Collab** using PyTorch
- Creates a customizable voice-over for the podcast using DeepGramAPI, **Pyttsx3**, and **gTTS**

Autonomous Video Content Generator | Personal | Troy, NY **June 2025 – August 2025**

- Scrapes Reddit stories using Python **Reddit API**, mapping subreddits and posts while tracking user profile data
- Converts stories into narrated Minecraft Youtube Shorts videos using **Pyttsx3** to voice-over, **FFmpeg** to add background visuals and music, **Whisper** run on **Docker** to transcribe voice to timestamped text, and **MoviePy** for custom hard subtitles

LEADERSHIP & INVOLVEMENT

Executive Board of RPI Union | Club and Organization Representative | Troy, NY **September 2025 – Present**

- Negotiated budgets for 5 RPI organizations as a liaison, established monthly initiative meetings between faculty and staff